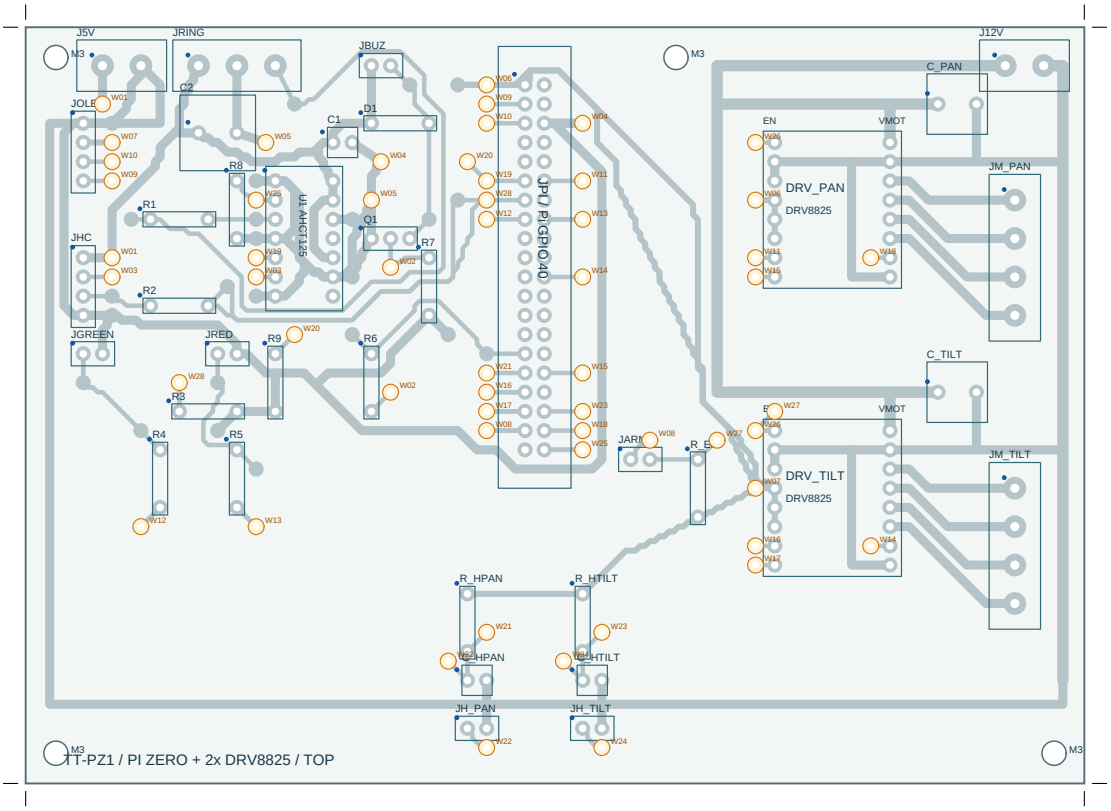
COMPONENT SIDE - 1:1 PLACEMENT

TT-PZ1 / 140 x 100 mm / exact scale at 100% print



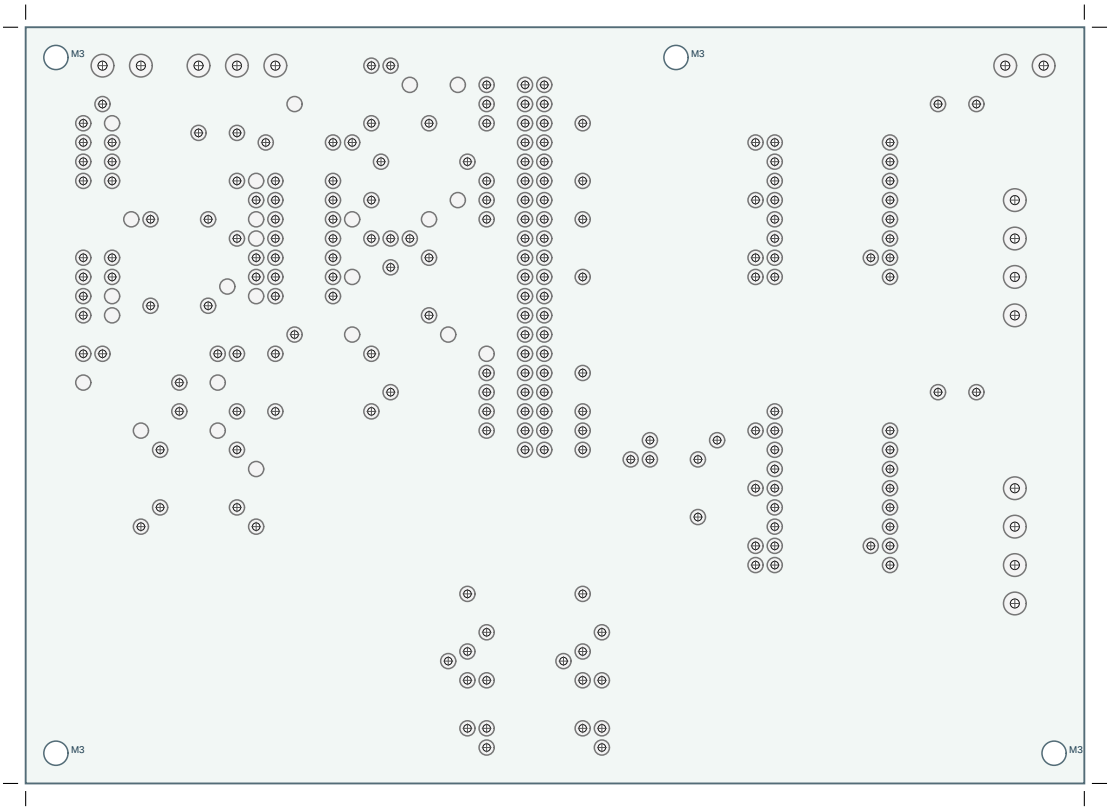
CALIBRATION: exactly 50 mm

Board outline 140 x 100 mm. A4 / Actual size / 100%.

All components on bare FR4 side. Copper is on the opposite side.
Blue dot = pin1. All copper pads round. JPI cable must be pin1-to-pin1.
DRV: EN upper-left, VMOT upper-right. U1 notch faces the board top.
Capacitor pin1 positive. D1 pin1 cathode. Q1 pads E-B-C left-to-right.
Keep JARM open until STEP/DIR software and homing are ready.

DRILL GUIDE - COMPONENT SIDE

TT-PZ1 / 140 x 100 mm / exact scale at 100% print



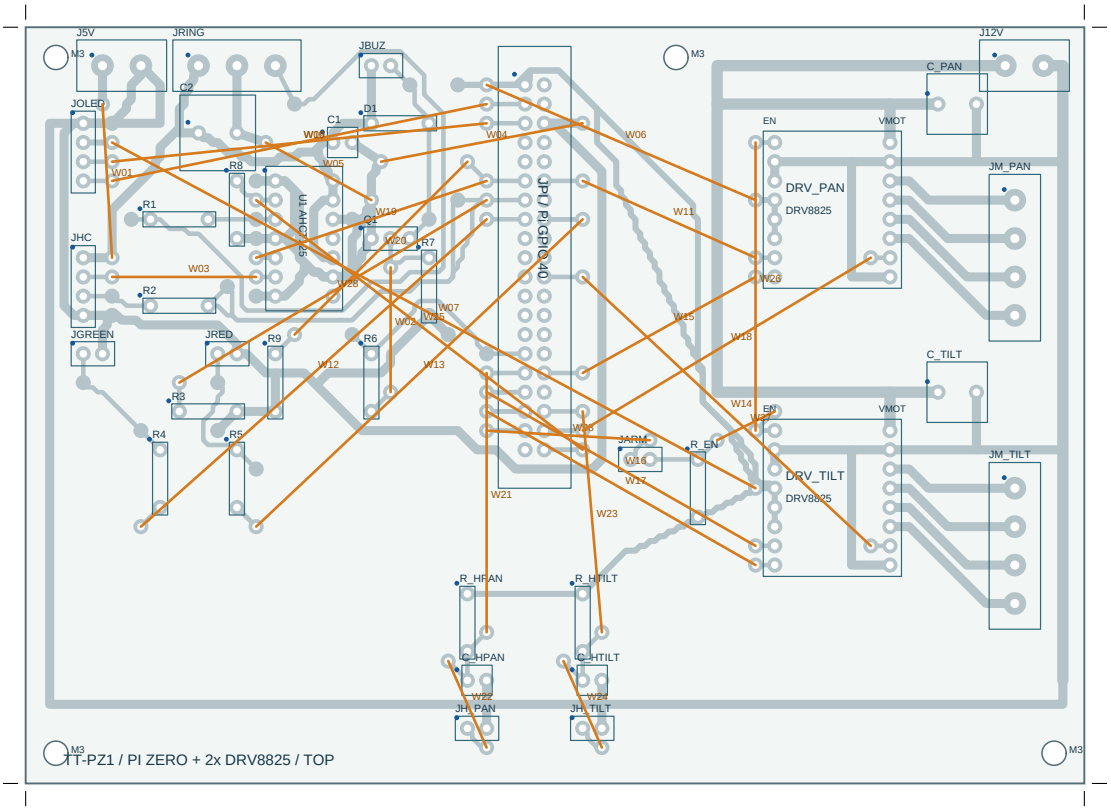
CALIBRATION: exactly 50 mm

Board outline 140 x 100 mm. A4 / Actual size / 100%.

Header / component / jumper holes: 1.0 mm.
Screw terminals: 1.2 mm. Mounts: 3.2 mm. Solid test pads: DO NOT DRILL.
Coordinates in pads.csv are millimetres from TOP-LEFT, component-side view.
Check footprint with the real part; form radial capacitor leads to 5.08 mm.
No plated holes required. Solder all pads on copper side.

INSULATED WIRE JUMPERS - TOP SIDE

TT-PZ1 / 140 x 100 mm / exact scale at 100% print



CALIBRATION: exactly 50 mm

Board outline 140 x 100 mm. A4 / Actual size / 100%.

Orange lines are insulated WIRE, not copper traces.
Each Wxx connects its two dedicated pads. Follow jumpers.csv endpoints.
Route around sockets/components as needed; cut length includes 20 mm slack.
AWG22 for GND / 5V / 12V; AWG26 for logic. Insulate every crossing.
Install wires before sockets; retain access to module trimpots and heatsinks.

WIRE SCHEDULE - COMPONENT SIDE

Pad coordinates from top-left. Dedicated pads already included in copper master.

REF	NET	PAD 1 (x,y mm)	PAD 2 (x,y mm)	CUT mm
W01	5V_AUX	11.43, 30.48	10.16, 10.16	40.4
W02	BUZZ_B	48.26, 31.75	48.26, 48.26	36.5
W03	TRIG_5V	30.48, 33.02	11.43, 33.02	39.0
W04	GND	46.99, 17.78	73.66, 12.70	47.1
W05	GND	31.75, 15.24	45.72, 22.86	35.9
W06	3V3	96.52, 22.86	60.96, 7.62	58.7
W07	3V3	11.43, 15.24	96.52, 60.96	116.6
W08	ENABLE_GPIO	82.55, 54.61	60.96, 53.34	41.6
W09	SDA	11.43, 20.32	60.96, 10.16	70.6
W10	SCL	11.43, 17.78	60.96, 12.70	69.8
W11	PAN_STEP	96.52, 30.48	73.66, 20.32	45.0
W12	GREEN_GPIO	15.24, 66.04	60.96, 25.40	81.2
W13	RED_GPIO	30.48, 66.04	73.66, 25.40	79.3
W14	TILT_FAULT	111.76, 68.58	73.66, 33.02	72.1
W15	PAN_DIR	96.52, 33.02	73.66, 45.72	46.2
W16	TILT_STEP	96.52, 68.58	60.96, 48.26	61.0
W17	TILT_DIR	96.52, 71.12	60.96, 50.80	61.0
W18	PAN_FAULT	111.76, 30.48	73.66, 53.34	64.4
W19	TRIG_3V3	30.48, 30.48	60.96, 20.32	52.1
W20	TRIG_3V3	35.56, 40.64	58.42, 17.78	52.3
W21	HOME_PAN	60.96, 80.01	60.96, 45.72	54.3
W22	HOME_PAN	60.96, 95.25	55.88, 83.82	32.5
W23	HOME_TILT	76.20, 80.01	73.66, 50.80	49.3
W24	HOME_TILT	76.20, 95.25	71.12, 83.82	32.5
W25	RING_3V3	30.48, 22.86	73.66, 55.88	74.4
W26	DRV_EN	96.52, 53.34	96.52, 15.24	58.1

WIRE SCHEDULE - COMPONENT SIDE

Pad coordinates from top-left. Dedicated pads already included in copper master.

REF	NET	PAD 1 (x,y mm)	PAD 2 (x,y mm)	CUT mm
W27	DRV_EN	91.44, 54.61	99.06, 50.80	28.5
W28	ECHO_3V3	20.32, 46.99	60.96, 22.86	67.3